

SEMA 4.x EC Linux GPIO Interrupt User Guide

Supported Hardware

1. Express-TL
2. cExpress-TL
3. Express-RLP
4. cExpress-RLP
5. cExpress-MTL
6. COM-HPC-cBLS
7. Express-PTL

Supported Operating Systems

The EC GPIO interrupt driver has been tested on the following operating systems:

1. Ubuntu 20, 22, 24
2. Redhat 9.5
3. Cent OS 10
4. Debian 12
5. SUSE Linux 15-SP6
6. Fedora 41

Note: cExpress-MTL, COM-HPC-cBLS & Express-PTL works only with Linux kernel 6.0 and above.

Load the SEMA drivers

1. Extract the SEMA 4.x EC zip package.
2. Open a terminal in the directory containing the SEMA source and run the command **“sudo make install”** to compile and install the SEMA drivers.
3. To load the SEMA drivers, enter the command:
“sudo modprobe -a adl-ec adl-ec-boardinfo adl-ec-vm adl-ec-wdt adl-ec-hwmon adl-ec-nvmem adl-ec-bklight adl-ec-i2c adl-ec-gpio adl-ec-nvmem-sec”

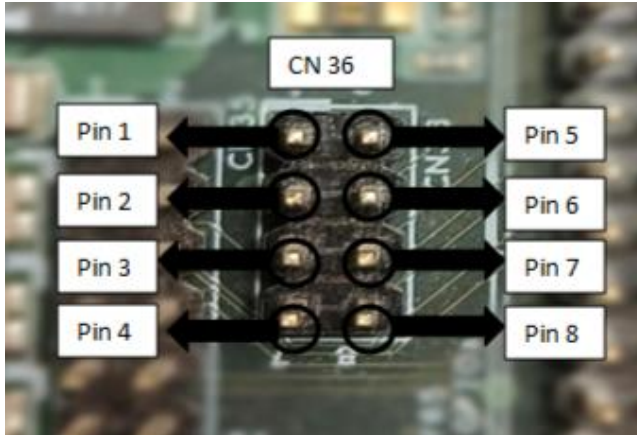
Loading the EC GPIO interrupt driver

1. Execute **“sudo make install_irq”** to compile and install the driver in the terminal.
2. Next, load the driver and pass the trigger type as an argument.
 - a. For Edge triggers:
sudo modprobe adl-ec-gpio-irq trigger_type=edge
 - b. For Low Level triggers:
sudo modprobe adl-ec-gpio-irq trigger_type=low-level
 - c. For High Level triggers:
#sudo modprobe adl-ec-gpio-irq trigger_type=high-level
3. To change the trigger type after the driver is loaded, unload the driver using the following command **“sudo modprobe -r adl-ec-gpio-irq”** and reload the driver with the required trigger type as the argument.

Testing the GPIO interrupts using SEMA

Set up the hardware

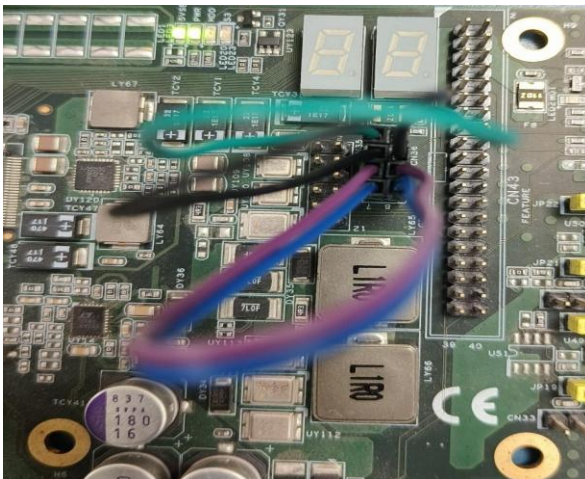
1. The carrier board contains four GPI pins and four GPO pins on CN36.



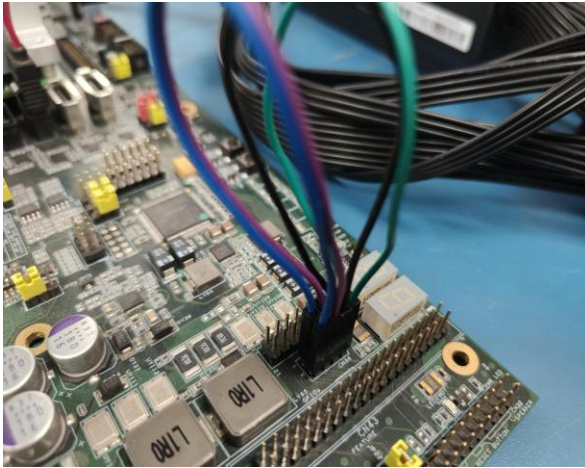
2. In CN36 pin configuration (default state)

Pin 1 – GPIO (INPUT)
Pin 2 – GPIO1 (INPUT)
Pin 3 – GPIO2 (INPUT)
Pin 4 – GPIO3 (INPUT)
Pin 5 – GPO0 (OUTPUT)
Pin 6 – GPO1 (OUTPUT)
Pin 7 – GPO2 (OUTPUT)
Pin 8 – GPO3 (OUTPUT)

3. Connect GPIO0 to GPO0, GPIO1 to GPO1, GPIO2 to GPO2 and GPIO3 to GPO3.



4. Pins 1 to 4 are GPI pins, and pins 5 to 8 are GPO pins.



Testing the EC GPIO interrupts

Edge Interrupts

To test the Edge interrupts:

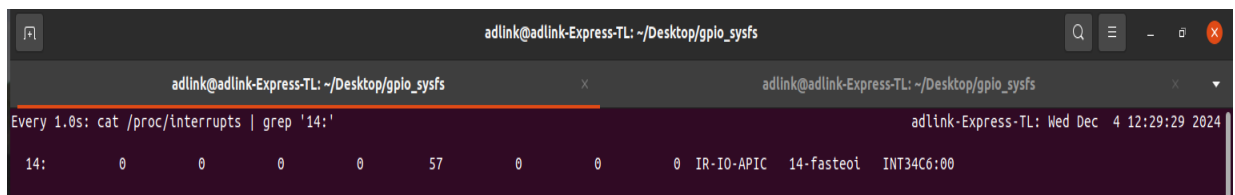
1. Continuously toggle any one of the output pins between high and low states.
 - i. Run the SEMA commands, for example, to change the GPO0 pin from high to low, enter the command “**sudo semautl /g set_level 5 0**”.
 - ii. Execute “**sudo semautl /g set_level 5 1**”, to change the GPO0 pin from low to high.
2. To observe the total number of interrupts generated, enter the command **# sudo watch -n 1 “cat /proc/interrupts | grep ‘14:’**”, and you will get the output shown in the image below. In this example, 57 represents the total number of GPIO interrupts generated.

```
adlink@adlink-Express-TL: ~/Desktop/gpio_sysfs
adlink@adlink-Express-TL: ~/Desktop/gpio_sysfs
Every 1.0s: cat /proc/interrupts | grep '14:'
14: 0 0 0 0 57 0 0 0 IR-IO-APIC 14-fasteoi INT34C6:00
adlink-Express-TL: Wed Dec 4 12:29:29 2024
```

Low-level interrupts

To test the Low-level interrupts:

1. All the GPO pins must be set to high initially. To do so, you can run the following commands.
 - a. **“sudo semautil /g set_level 5 1”** to set GPO0 high.
 - b. **“sudo semautil /g set_level 6 1”** to set GPO1 high.
 - c. **“sudo semautil /g set_level 7 1”** to set GPO2 high.
 - d. **“sudo semautil /g set_level 8 1”** to set GPO3 high.
2. To trigger a low-level interrupt, change any one of the GPO pins to a low state. For example, to change the GPO1 (pin 6) to low, enter the command **“sudo semautil /g set_level 6 0”**.
3. Low-level interrupts will continue to be generated until the pin is set back to a high state. For example, **“sudo semautil /g set_level 6 1”**.
4. To observe the total number of interrupts generated, enter the command **# sudo watch -n 1 “cat /proc/interrupts | grep ‘14:’”**, and you will get the output shown in the image below. In this example, 57 represents the total number of GPIO interrupts generated.



The image shows a terminal window with the command `cat /proc/interrupts | grep '14:'` being executed repeatedly. The output shows the number of interrupts for each pin, with the total for pin 14 being 57. The terminal window has a title bar that reads "adlink@adlink-Express-TL: ~/Desktop/gpio_sysfs".

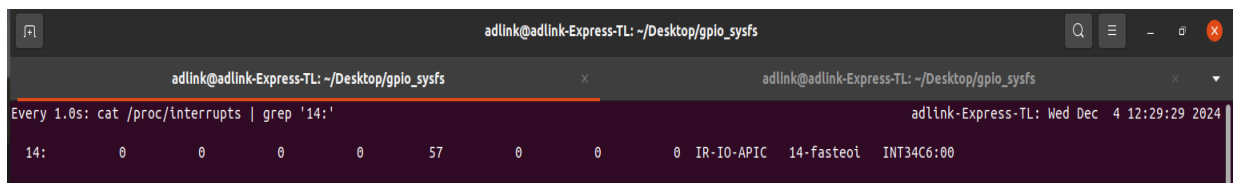
```
adlink@adlink-Express-TL: ~/Desktop/gpio_sysfs
Every 1.0s: cat /proc/interrupts | grep '14:'                                adlink-Express-TL: Wed Dec  4 12:29:29 2024
14:          0          0          0          0          57          0          0          0 IR-IO-APIC 14-fasteoi INT34C6:00
```

High-level interrupts

To test the High-level interrupts:

1. All the GPO pins must be set to low initially. To do so, you can run the following commands.
 - a. **“sudo semautil /g set_level 5 0”** to set GPO0 low.
 - b. **“sudo semautil /g set_level 6 0”** to set GPO1 low.
 - c. **“sudo semautil /g set_level 7 0”** to set GPO2 low.

- d. **“sudo semautil /g set_level 8 0”** to set GPO3 low.
2. Next to trigger a high-level interrupt, change the level of any one of the GPO pins to a high level. For example, to change the GPO2 (pin 7) to high, enter the command **“sudo semautil /g set_level 7 1”**.
3. High-level interrupts will continue to be generated until the pin is set back to a low state. For example, **“sudo semautil /g set_level 7 0”**.
4. To observe the total number of interrupts generated, enter the command **# sudo watch -n 1 “cat /proc/interrupts | grep ‘14:’”**, and you will get the output shown in the image below. In this example, 57 represents the total number of GPIO interrupts generated.



```
adlink@adlink-Express-TL: ~/Desktop/gpio_sysfs
Every 1.0s: cat /proc/interrupts | grep '14:'                                adlink-Express-TL: Wed Dec 4 12:29:29 2024
14:          0          0          0          0          57          0          0          0 IR-IO-APIC 14-fasteoi INT34C6:00
```

Note: Do not copy and paste the commands into the Linux terminal. Please type the commands manually before executing them.